

AWS Migration Business Case

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Executive Summary

This assessment analyzed **18 servers and 34 storage volumes** for AWS migration. The analysis reveals an estimated **annual savings of \$70,309 (76.9%)** by migrating to AWS, with total recommended AWS cost of **\$21,135** annually compared to **\$91,444** for on-premises infrastructure.

Assessment Scope

- **Compute**
 - **EC2:** 18 servers, 70 vCPUs, 216.0 GB memory
- **Storage**
 - **Block Storage (EBS):** 34 volumes, 4305.00 GiB provisioned, 2152.50 GiB used
- **On-Premises**
 - **On-Premises:** 18 Virtual Servers, 34 Server Volumes, 4.20 TiB storage
- **Excluded from Assessment Scope**
 - **Storage:** 142 excluded
 - 142 — attached to excluded servers

Please refer to the Assessment Issues sheet in the analysis workbook for additional details on issues found with inventory.

Strategic Benefits

Strategic Benefit	Business Impact
Cost Optimization	Estimated annual savings of \$70,309 (76.9%) with 3-Year Reserved Instances vs on-demand
Infrastructure Modernization	Migration from aging bare metal to cloud-native, right-sized EC2 instances
Operational Efficiency	Elimination of hardware maintenance, power, cooling, and data center overhead costs
Enhanced Scalability	On-demand resource scaling to meet changing business requirements
Improved Reliability	AWS 99.99% SLA with built-in redundancy and disaster recovery capabilities

Insights

Financial Summary

Cost Component	AWS - On Demand	AWS - 1 Year No Upfront	AWS - 3 Year No Upfront
Compute	\$24,514	\$19,152	\$16,034
Storage	\$2,175	\$2,175	\$2,175
Network	\$1,004	\$1,004	\$1,004
Business Support	\$2,769	\$2,233	\$1,921
Annual Total	\$30,463	\$24,564	\$21,135
Savings vs On-Demand	-	\$5,899 (19.4%)	\$9,328 (30.6%)

Next Steps

1. **Engage Specialists:** [Engage a specialist](#) if you want further support and guidance on your migration journey.
2. **Perform wave planning and migration:** Utilize the AWS Transform VMware job capabilities to perform comprehensive application assessment through dependency mapping, create migration waves, and execute the VMware-to-EC2 migrations
3. **Engage AWS Migration Acceleration Program:** Apply for the [AWS MAP program](#) to potentially offset migration costs and accelerate your cloud journey through expertise from AWS Professional Services, a global partner ecosystem, and AWS investment.

On-Premises Assessment

Scope

Metric	Value
Total Servers	18 Virtual Servers, 34 Server Volumes

Cost Breakdown

Cost Category	Annual Cost
Compute	\$53,327
Storage	\$36,145
Network	\$1,972
Total	\$91,444

Assumptions

- TCO Duration: 5 years
- AWS Region: us-east-1
- Compute costs include: server and hardware costs (based on industry benchmarks for CPU and Memory), software licensing costs (OS), facility costs
- Storage costs include: server volumes
- Network costs include: software, hardware, and bandwidth

Compute Assessment

Cost Summary

Cost Category	On Demand	1 Year No Upfront	3 Year No Upfront
Compute	\$24,514	\$19,152	\$16,034
Network	\$1,004	\$1,004	\$1,004
Total	\$25,519	\$20,156	\$17,039

EC2 Migration

Scope

Metric	Value
Servers Analyzed	18 servers
Capacity Analyzed	70 vCPUs, 216.0 GB memory
Windows Servers	16
Linux Servers	2

Cost Breakdown

Cost Category	On-Demand	1-Year Reserved Instances	3-Year Reserved Instances
Compute	\$24,514	\$19,152	\$16,034
Network	\$1,004	\$1,004	\$1,004
Total	\$25,519	\$20,156	\$17,039

Key Findings

1. Analyzed 18 active servers for EC2 migration
2. Successfully mapped 18 servers to EC2 instances
3. Recommended pricing: 3-Year Reserved Instances
4. Right-sizing applied: 25% CPU, 60% memory utilization

Recommendations

1. **Implement Reserved Instance Strategy:** Purchase 3-Year Reserved Instances for predictable workloads to achieve savings over on-demand pricing.
2. **Right-Sizing Applied:** Instance sizes adjusted based on 25% CPU, 60% memory utilization to optimize costs.

Assumptions

- Region: us-east-1
- Tenancy: Shared
- Pricing model: 3-Year Reserved Instances (No Upfront)
- Architecture: x86 (Intel and AMD)
- Right-sizing: 25% CPU, 60% memory utilization
- Network cost: 10.0% of 1-Year Reserved Instances base instance cost

Storage Assessment

Scope

Metric	Value
EBS Storage	34 volumes
EBS Capacity	4305.00 GiB provisioned, 2152.50 GiB used

Cost Breakdown

Storage Service	Annual Cost
EBS	\$2,175
Total	\$2,175

Key Findings

1. Only cost-optimized strategy is included in the assessment because 1-to-many and cost-optimized strategies are identical. This may be due to missing peak IOPS and throughput data in the input inventory which results in using default values from configuration
2. All 34 virtual storage volumes successfully mapped to AWS EBS GP3 (General Purpose SSD)
3. Total annual cost of \$2,175.12 includes 34 EBS GP3 volumes with 2,266 GB total capacity
4. Storage utilization at 50% (2,153 GB used out of 4,305 GB provisioned) based on default capacity percentage
5. Assessment includes 1,020 GB of boot volume capacity (34 virtual machines × 30 GB each)

Assumptions

- Assessment performed for AWS region: us-east-1
- All volumes mapped to None as per configuration override

Disclaimer

This report provides an estimate of fees in USD, and savings based on certain information you provide. Fee estimates do not include any taxes that might apply. Your actual fees and savings depend on a variety of factors, including your actual usage of AWS services, which may vary from the estimates provided in this report. Additional configurations are available on request by engaging a specialist.